

Lesson 6: Algebraic Expressions

Writing Algebraic Expressions

An **algebraic expression** uses numbers, operations, and variables to show number relationships. **Variables** are letters (such as x and y) that represent unknown numbers. Each time a letter is used within the same expression, it represents the same number.

To solve algebra problems, you will need to be able to translate number relationships described in words into algebraic expressions. Study the following examples.

Key Ideas

- Algebraic expressions show mathematical relationships using numbers, symbols, and variables.
- Variables are letters that take the place of unknown numbers.

Algebraic Expressions in Words	In Symbols
the product of 5 and a number	$5x$
a number decreased by 12	$x - 12$
the sum of 3 and the square of a number	$3 + x^2$
6 less than the quotient of a number and 2	$\frac{x}{2} - 6$
one-half a number increased by 15	$\frac{1}{2}x + 15$
4 times the difference of -3 and a number	$4(-3 - x)$
a number less another number	$x - y$
10 less the square root of the sum of a number and 3	$10 - \sqrt{x + 3}$

To do well on algebra questions on the GED® *Mathematical Reasoning Test*, you must be able to translate a common life situation into mathematical symbols. You will use this skill to write equations and functions, to apply formulas, and to solve word problems.

GED® TEST TIP

To check whether the expression you have chosen is correct, substitute easy numbers into the expression and complete the operations. Then see if the result is reasonable for the situation.

Example 1: Kyle processes sales for an online bookstore. The shipping and handling on an order is equal to 4 percent of the total cost of the order plus \$0.95 per book. If c represents total cost, in dollars, and n represents the number of books in an order, which of the following expressions could be used to find the shipping and handling for an order?

- (1) $\frac{4}{100}nc + 0.95$
- (2) $(0.04 + 0.95)n + c$
- (3) $0.04c + 0.95n$

This kind of problem is called a **setup problem**. You need to recognize the correct way to find the shipping and handling based on the total cost and number of items. The relationship is described in the second sentence.

$$\begin{aligned}\text{shipping and handling} &= 4\% \text{ of total cost } (c) \text{ plus } \$0.95 \text{ per book } (n) \\ &= 0.04c + 0.95n\end{aligned}$$

The correct answer is option (3), $0.04c + 0.95n$.

Practice 6.1

A. Write an algebraic expression for each description. Use the variables x and y .

- a number decreased by 7
- the product of 3 and the square of a number is increased by that number
- the product of 8 and a number is decreased by 10
- the difference of -3 multiplied by a number and the product of 2 and another number
- 5 less than the quotient of 10 and a number
- the sum of -8 and the product of 7 and a number
- the sum of 16 times a number and the number is decreased by 3 times another number
- a number squared plus the number raised to the fourth power
- the square of a number plus the quotient of 4 and 7

- 6 subtracted from the sum of 15 and the square root of a number $\rightarrow = \sqrt{x} + 15 - 6$
- a number less the sum of another number and 13
- the square of the sum of a number and 6
- 17 less the sum of 2 times a number plus another number
- a number increased by the quotient of 24 and the number
- the difference of the product of 2 and a number and 15
- 4 times the difference of two different numbers $\rightarrow = 4(x - y)$
- 5 multiplied by the difference of a number squared and 3
- the product of a number and the difference of 11 and the square root of 100 $\rightarrow = x(11 - \sqrt{100}) = x(11 - 10) = x$

B. Choose the one best answer to each question.

19. A minor-league baseball team is giving a local charity the sum of \$1500 and \$0.50 for each ticket over 2000 sold for one game. Let x represent the number of tickets sold. If the team sells more than 2000 tickets, which of the following expressions could be used to find the amount of the donation?

- $\$1500 + \$0.50x$
- $\$1500 + \$0.50(2000 - x)$
- $\$1500 + \$0.50(x - 2000)$
- $\$1500(2000 - x)(\$0.50)$

20. The sum of 3 times a number and 4 times a second number is divided by the sum of 2 and a third number. Which of the following expressions represents this series of operations?

- $(3x + 4y) \div (2 + z)$
- $3x + 4y \div (2 + z)$
- $3x + 4y \div 2 + z$
- $(3x + 4y) \div 2z$

Question 21 refers to the following information.

Appliance City employees earn an hourly wage plus commission. Wage options are shown below.

Option	Hourly Wage	Commission on Sales
A	\$7.50	1%
B	\$6.00	3%

21. Chandra is paid under Option B. If h represents the number of hours worked and s represents Chandra's total sales, in dollars, which of the following expressions could be used to find her weekly pay?

- $6 + h + 0.03s$
- $6h + 0.03s$
- $6s + 0.03h$
- $0.03(h)(s)$

Answers and explanations begin on page 672.

Simplifying and Evaluating Expressions

Simplifying an expression means to perform all the operations you can within an expression. When working with variables, you must remember an important rule: you can add or subtract like terms only.

A **term** is a number, a variable, or the product or quotient of numbers and variables. A term cannot include a sum or a difference.

Examples: $5x$ $3y^2$ 13 x^3 $\frac{x}{2}$

Like terms have the same variable raised to the same power. For example, $3x^2$ and $5x^2$ are like terms. $8y$ and $4y$ are also like terms. However, $6x$ and $2x^2$ are not like terms because the variables are not raised to the same power.

To simplify an expression, combine like terms.

Example 2: Simplify $2x - 5 + 4x^2 - 8 + 6x$.

Combine like terms. It is customary to write the term with the greatest exponent first and to continue in descending order.

$$\begin{aligned} 2x - 5 + 4x^2 - 8 + 6x \\ = (2x + 6x) + (-5 + -8) + 4x^2 \\ = 8x + (-13) + 4x^2 \\ = 4x^2 + 8x - 13 \end{aligned}$$

The **distributive property** allows you to remove grouping symbols to simplify expressions. We can state the distributive property using symbols.

$$a(b + c) = ab + ac \quad \text{and} \quad a(b - c) = ab - ac$$

In other words, each term inside the parentheses is multiplied by the term outside the parentheses, and the results are added or subtracted depending on the operation inside the parentheses. Example 3 applies the distributive property.

Example 3: Simplify $4x - 3(x + 9) + 15$.

1. Change subtracting to adding a negative number.
2. Use the distributive property. Multiply -3 by each term in the parentheses.
3. Combine like terms.
(NOTE: $1x$ means x .)

$$\begin{aligned} 4x - 3(x + 9) + 15 \\ = 4x + -3(x + 9) + 15 \\ = 4x + (-3x) + (-3)(9) + 15 \\ = 4x + (-3x) + (-27) + 15 \\ = [(4x + (-3x))] + (-27 + 15) \\ = \underline{x - 12} \end{aligned}$$

Evaluating an expression means finding its value. To evaluate an expression, substitute a given number for each variable. Follow the order of operations.

Example 3: Find the value of the expression $\frac{3x + 2y}{4}$ when $x = 6$ and $y = 5$.

1. Replace the variables with the values given in the problem.
2. Perform the operations above the fraction bar. Then divide.

$$\begin{aligned} \frac{3x + 2y}{4} &= \frac{3(6) + 2(5)}{4} \\ \frac{3(6) + 2(5)}{4} &= \frac{18 + 10}{4} = \frac{28}{4} = 7 \end{aligned}$$

NOTE: To remove parentheses from an operation that follows a minus sign, imagine that the parentheses are preceded by 1. Then use the distributive property.

$$\begin{aligned} -(2x + 3) \\ = -1(2x + 3) \\ = -1(2x) + (-1)(3) \\ = -2x + (-3) \text{ or } -2x - 3 \end{aligned}$$

Practice 6.2

A. Simplify.

1. $5 + x^2 - 3 + 3x$
2. $2y + 5 + 17y + 8$
3. $3x - 6(x - 9)$
4. $6x^3 + 4 + 2x^2(15) + x^2$
5. $4(y + 8) + 3(y - 6)$

$$6. 5 - (x - 3) + 4x$$

$$7. 16x + 6(x - 2)$$

$$8. 5y^2 + 4 - 3y^2 + 5 + y$$

$$9. -3(x + 3) - 2(x + 4)$$

$$10. 5x - (x + 4) - 3$$

$$= 4y + 32 + 3y - 18$$

$$= 7y + 14$$

$$= -3x - 9 - 2x - 8$$

$$= -5x - 17$$

B. Evaluate each expression as directed.

$$11. \text{ Find the value of } 6(x + 2) + 7 \text{ when } x = 2.$$

$$12. \text{ Find the value of } 3x^2 + 3(x + 4) \text{ when } x = 3.$$

$$13. \text{ Find the value of } \frac{(x+y)^2}{2} - 10 \text{ when } x = 2 \text{ and } y = 4.$$

$$14. \text{ Find the value of } y^2 + 16 - (y - 5)^2 \text{ when } y = 3.$$

$$15. \text{ Find the value of } 8x + 9y - (2x + y) \text{ when } x = 4 \text{ and } y = 6.$$

$$16. \text{ Find the value of } x^2 + 3y - 4 + 2(x - z) \text{ when } x = 7, y = 5, \text{ and } z = -3.$$

$$17. \text{ Find the value of } (14 - x)^2 + 20\sqrt{x} \text{ when } x = 9.$$

$$18. \text{ Find the value of } \frac{3(2x - y)}{3} + 6(y - 5) \text{ when } x = -2 \text{ and } y = 3.$$

$$19. \text{ Find the value of } x^2 - (x^3 + 3) \text{ when } x = -2.$$

$$20. \text{ Find the value of } 30x + 2 + 2y^2 - 3(x - 2)^2 \text{ when } x = 1 \text{ and } y = 4.$$

$$= \frac{3[2(-2) - 3]}{3} + 6(3 - 5)$$

$$= \frac{3(-7)}{3} + (-12)$$

$$= -7 - 12$$

$$= -19$$

C. Choose the one best answer to each question.

$$21. \text{ Which of the following expressions is equal to } 3x^2 + 3(x - 3) + x + 10?$$

- A. $x^2 + 9x + 1$
- B. $3x^2 + 4x + 19$
- C. $3x^2 - 2x + 19$
- D. $3x^2 + 4x + 1$

$$22. \text{ Given the expression } 4x^2 - 3(y + 6), \text{ which of the following values for } x \text{ and } y \text{ will result in a value of } -11?$$

- A. $x = 2, y = 3$
- B. $x = -2, y = 4$
- C. $x = -1, y = 2$
- D. $x = 1, y = 0$

Question 23 refers to the following information.

Temperature Conversion Formulas	
To convert Fahrenheit (F) to Celsius (C)	$C = \frac{5}{9}(F - 32)$
To convert Celsius (C) to Fahrenheit (F)	$F = \frac{9}{5}C + 32$

$$23. \text{ If the temperature is } 68^\circ \text{ Fahrenheit, what is the temperature in Celsius?}$$

- A. 20°
- B. 36°
- C. 154.4°
- D. 180°

Answers and explanations begin on page 672.

Factors and Multiples

When you multiply two numbers to create a product, the numbers you multiply are **factors**. The pairs of factors that can be multiplied to produce a whole number can be easily listed in a T-chart.

Example 1: What are all the factors of 10?

1. Make a T-chart:

10		
1	10	$1 \times 10 = 10$
2	5	$2 \times 5 = 10$
(-1)	(-10)	$(-1) \times (-10) = 10$
(-2)	(-5)	$(-2) \times (-5) = 10$

2. Count the factors in the T-chart. The number 10 has eight distinct factors. In order from least to greatest, they are **-10, -5, -2, -1, 1, 2, 5, 10**.

Prime numbers are positive whole numbers with only two positive factors, 1 and the number itself. Examples of prime numbers are 2, 3, 5, 7, 11, and 13. (Note that 1 is *not* a prime number.)

For example, 10 is not a prime number because, as just discovered, it has four positive factors: 1, 2, 5, and 10. However, the numbers 2 and 5 are prime numbers. The only positive factors of 2 are 1 and 2, and the only positive factors of 5 are 1 and 5. Thus, the **prime factorization** of 10 is 2×5 . A number's prime factorization may include more than two factors, as in the next example.

Example 2: What is the prime factorization of 12?

1. Break 12 into its prime factors.

$$12 = 4 \times 3 = 2 \times 2 \times 3$$

Prime factors can be used to find two values that may be useful in simplifying expressions and solving equations.

The **greatest common factor (GCF)** of two integers is the largest number that divides evenly into both integers. To find the GCF, break down both integers into their prime factorizations, find the prime factors they have in common, and multiply those prime factors. This is handy when using the distributive property to rewrite two or more terms as one.

Example 3: What is the greatest common factor of 40 and 140?

1. Identify the prime factors of each number.

$$40 = 4 \times 10 = (2 \times 2) \times (2 \times 5) = 2 \times 2 \times 2 \times 5$$

$$140 = 10 \times 14 = (2 \times 5) \times (2 \times 7) = 2 \times 2 \times 5 \times 7$$

2. See what prime factors the two numbers have in common.
Both integers share two 2s and one 5.

3. Multiply these common factors.

The greatest common factor is $2 \times 2 \times 5 = 20$. This is the largest number that divides evenly into both 40 and 140. Note that you could rewrite the expression $40 + 140$ as $20(2 + 7)$.

Key Ideas

- The prime factorization of an integer is the prime numbers that can be multiplied together to equal that integer.
- The greatest common factor of two integers is the largest integer that will divide evenly into both.
- The least common multiple of two integers is the smallest number that both integers will divide evenly into.

GED® TEST TIP

When you need to add or subtract fractions with different denominators and the denominators are large, instead of multiplying the denominators to find a common denominator, use prime factorization to find their least common multiple.

The **least common multiple (LCM)** of two integers is the smallest number that both integers divide into. To find the LCM, break down both integers into their prime factorizations, select each prime factor the most times it occurs, and multiply all of the prime factors you have selected.

Example 4: What is the least common multiple of 20 and 30?

1. Identify the prime factors of each number

$$20 = 2 \times 2 \times 5 \quad 30 = 2 \times 3 \times 5$$

2. Select each factor the most times it occurs. The factor 2 appears twice in 20, so select 2×2 . The factor 3 appears once in 30, so select 3. And the factor 5 appears once in both 20 and 30, so select it once.

3. Multiply these factors.

The least common multiple is $2 \times 2 \times 3 \times 5 = 60$. This is the smallest number that both 20 and 30 will divide into. Note that if you needed to solve $\frac{7}{20} + \frac{7}{30}$, you could use a common denominator of 60: $\frac{21}{60} + \frac{14}{60} = \frac{35}{60} = \frac{7}{12}$.

Practice 6.3

A. Find the prime factorization of the following numbers.

1. 6 2. 21 3. 36 4. 54 5. 70

B. For each pair of numbers below, find the greatest common factor and the least common multiple.

6. 6, 9 7. 45, 75 8. 50, 125

C. Solve.

9. $30 + 75 =$

- A. $5(3 + 7)$
B. $10(6 + 5)$
C. $15(2 + 5)$
D. $150(5 + 2)$

10. $\frac{11}{75} - \frac{2}{45} =$

- A. $\frac{23}{225}$
B. $\frac{43}{225}$
C. $\frac{3}{10}$
D. $\frac{3}{5}$

Answers and explanations begin on page 673.

Lesson 7: Expressions and Calculator Skills

When you take the GED® *Mathematical Reasoning Test*, you may use either a handheld or an online version of a scientific calculator (TI-30XS MultiView™) to use on the second part of the test items. This calculator, like most scientific calculators, uses algebraic logic, which means that it follows the order of operations that you saw on page 324.

You need to practice using a scientific calculator with algebraic logic. You can find out whether your calculator uses algebraic logic by running this simple test.

Press: $4 \times 3 x^2 \text{enter}$. (Your calculator may have an equal sign instead of an *enter* button.)

If the display reads **36**, your calculator uses algebraic logic.

If the display reads **144**, your calculator does not use algebraic logic. You should find another calculator to practice for the *Mathematical Reasoning Test*.

You can use a calculator to evaluate an expression that contains several operations.

Example 1: Find the value of the expression $2x^2 + 3x - 5$ when $x = -4$.

When you come to the variable x , enter -4 by pressing $(-)$ 4. The $(-)$ key is called the **change sign key**.

Press: $2 \text{ () } (-) 4 \text{ () } x^2 + 3 \times (-) 4 - 5 \text{ enter}$.

The right side of the display reads 15.

The value of the expression is **15**.

Expressions sometimes contain grouping symbols to show a different order of operations. You can enter grouping symbols on a scientific calculator. On the TI-30XS MultiView™, the grouping symbols $(\text{)}$ and $(\text{)}$ are found above the $\frac{\square}{\square}$ and $\frac{\square}{\square}$, respectively. When you enter the left, or open, parenthesis, $(\text{)}$, the calculator waits until you enter the right, or closing, parenthesis,) , before it calculates what is inside the parentheses.

Example 2: Find the value of the expression $2(x + 4) + \frac{5x}{3}$ when $x = 6$.

Press: $2 \times (\text{) } 6 + 4 \text{) } + 5 \times 6 \div 3 \text{ enter}$.

The right side of the display reads 30.

The value of the expression is **30**.

You can also use your calculator for only part of an expression.

Example 3: Find the value of the expression $\frac{3x + 6}{2} + \sqrt{225}$ when $x = 4$.

Substitute 4 for x in the first part of the expression and calculate the results by hand or using your calculator: $\frac{3(4) + 6}{2} = \frac{12 + 6}{2} = \frac{18}{2} = 9$.

Now use your calculator to find the square root of 225.

Press: $2\text{nd } x^2 225 \text{ enter}$. The right side of the display reads 15.

Add the results of the two steps: $9 + 15 = 24$.

Key Ideas

- ON THE *Mathematical Reasoning Test*, YOU WILL USE A SCIENTIFIC CALCULATOR THAT FOLLOWS THE ORDER OF OPERATIONS.
- USE THE *change sign* KEY TO ENTER A NEGATIVE NUMBER.
- USE THE *grouping symbol* KEYS TO CHANGE THE ORDER OF OPERATIONS.

GED® TEST TIP

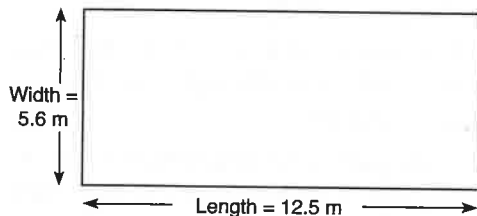
To check your work when using a calculator, enter each calculation twice. If the results are the same, you probably pressed the keys that you intended.

Practice 7

A. Use a calculator as needed to find the value of the expressions as directed.

- What is the value of $5x^2 - 3x + 5$ when $x = 2$?
- Find the value of $\sqrt{7}x + 2x$ to the nearest tenth when $x = 5$.
- If $x = -3$, what is the value of $7x^2 + 2x - 6$?
- What is the value of $\frac{1}{2}x + 15$ when $x = 3$?
- Find the value of $3(2x + 3 + y) - 14$ when $x = -2$ and $y = 9$.
- If $y = -3$, what is the value of $4y^3 + 2(y^2 - 4)$?
- What is the value of $2(x^2 + 6) + 3(x - 1)$ when $x = 5$?
- If $x = 4$ and $y = -4$, what is the value of the expression $6x^2 + 3y^2 + 2$?
- Find the value of $-2(x^3 + 3) + 16x + 2$ when $x = 2$.
- If $x = 7$, what is the value of the expression $7 + 3(x - 2) - 2x^2$?
- Find the value of $-(x + y) + 3(2z - y)$ if $x = -5$, $y = -7$, and $z = 4$.
- What is the value of the expression $x^2 - 7(3 - y) + 4$ when $x = 5$ and $y = 4$?
- If $x = -2$ and $y = 8$, what is the value of $x^2 + y - 6(y + 3)$?
- What is the value of $\sqrt{760 - 4x^2}$ to the nearest tenth when $x = 6$?
- If $x = 20$, what is the value of the expression $4(x + 7) - 3(x - 2)$?
- Find the value of $\frac{x(25 + 2x - y)}{-z}$ when $x = -3$, $y = 4$, and $z = -1$.
- What is the value of $5y^2 + 4x^2 - 6(x - y)$ when $x = 5$ and $y = -2$?
- Find the value of $3x^2 \cdot \frac{2(x - 3y)}{6}$ when $x = 6$ and $y = 1$.
- What is the value of the expression $(x^2 + 5)(x^2 - x + 2) - 3$ when $x = -4$?
- Find the value of $(x + y)(x - y)(2x + y)$ when $x = 9$ and $y = 4$.

B. Choose the one best answer to each question.



21. Jake has to buy enough fencing to enclose the rectangular garden shown above.

The formula for finding the perimeter of (or distance around) a rectangle is $P = 2l + 2w$, where P = perimeter, l = length, and w = width. Using the values from the drawing, what is the perimeter, in meters, of the garden?

- 18.1
- 36.2
- 70
- 140

22. If $x = -5$ and $y = 2$, which of the following expressions has the greatest value?

- $x + y$
- $-x + y$
- xy
- $-2xy$

23. What is the value of the expression $2 \div x^{-4}$ when $x = 2$?

- 32
- 16
- 8
- 16

Answers and explanations begin on page 673.

Lesson 8: Understand Polynomials

Key Ideas

- A polynomial is an expression with more than one term—any combination of coefficients, variables, exponents, and constants.
- Specific types of polynomials include monomials, binomials, and trinomials.

GED® TEST TIP

If there is no coefficient next to a variable, assume the coefficient is 1.

The GED® *Mathematical Reasoning Test* will expect you to understand polynomials. A **polynomial** is made up of terms, and since the prefix *poly-* means “many,” a polynomial is a term in which many parts are being combined. Each term in a polynomial may be made up of a combination of coefficients, variables, exponents, and constants.

Coefficients include numbers such as 4, -25 , or $\frac{1}{2}$ and will come before the variables. Coefficients can be negative, like -25 . If there is no coefficient next to a variable, assume the coefficient is 1. In the expression $x + y + z$, the coefficient in front of each variable is equal to 1.

Variables are letters such as x and y in polynomial terms. In the polynomial $45a^2b$, the variables are a and b .

Exponents, or powers, are a way of showing repeated multiplication. This is also known as raising a number or variable to a power. Polynomials must be raised to whole number exponents. There are no negative or fractional powers in polynomials. In the polynomial $6x^3y^4$, the exponents are 3 and 4. The **degree** of a polynomial with only one variable is the largest exponent of that variable. For example, in the expression $4x^3 + 3x^2 - 5$, the degree is 3 (the largest exponent of x .)

A **constant** is a number on its own that will always remain the same. It is not next to a variable. A constant can be either positive or negative, depending on whether it is being added or subtracted. In the expression $4x^3 + 3x^2 + 5$, the constant is 5. No matter what value you substitute for x , 5 will always be just 5.

Polynomials are built out of all of the components above, and there are different types of polynomials, including monomials, binomials, and trinomials. These are important to know.

- A **monomial** is a polynomial with only one term. An example of a monomial is $4xy^2$. This term is made up of a coefficient (4), two variables (x and y), and an exponent (the power of 2).
- A **binomial** is a polynomial with two terms. An example of a binomial is $3x + 2$. The first term is made up of a coefficient (3) and a variable (x), and the second term consists of a constant (2).
- A **trinomial** is a polynomial with three terms. An example of a trinomial is $4x + 3y^2 - 4$. The first term is made up of a coefficient (4) and a variable (x). The second term is made up of a coefficient (3), a variable (y), and an exponent (the power of 2). The third term is made up of a constant, and this time it is a negative number (-4).

A few types of numbers are not categorized as polynomials and cannot be present in polynomial terms. These include the following:

- Division by a variable, such as $\frac{3}{x+3}$ or $\frac{1}{x}$
- Negative exponents, such as $4xy^{-2}$ (exponents can only be 0, 1, 2, etc.)
- Fractional exponents, such as $3a^{\frac{1}{2}}$
- Variables inside radicals, such as $\sqrt{x}$

In a polynomial term, however, you can divide by a constant or have numbers inside a radical. For example, $\frac{4x}{12}$ is allowed, as is $\sqrt{10}$.

Practice 8

A. Identify whether each of the following is a monomial, binomial, or trinomial.

1. $25a$
2. $2xy^2z$
3. $x - 4$
4. $x^3 + y - 1$
5. $7y - 1$
6. $2x^4 + 3x + 4$
7. $\frac{x}{3}$
8. y^2
9. $g^2h^2i^2j^2$
10. $x^2 + x^2$
11. $x^2 + 14x + 3$
12. $x^2y + x^2 + y$
13. $g + h$
14. $\sqrt{49}$
15. $3x^2 - 5$

B. For each polynomial, identify the terms. Remember that a coefficient can be positive or negative.

16. $3x^4 - 2x^2 + 3$
17. $12a^2bc$
18. $3g - 4h$
19. $x^2 + y$
20. $-4a - 3b^2 + c$
21. 25
22. $x^2 + 3x - 7$
23. $\frac{3x}{8}$
24. $\sqrt{25}$
25. $\frac{x^2}{9}$
26. $49x^2y^2z^2$
27. $18y^2 - 4y^2 + 8$
28. $3h - 4$
29. $x^2 - x + y^2 - 2$
30. $ab + ab^2 + b^2 - 4$

C. Choose the one best answer to each question.

31. What is the sum of the exponents in the expression $4x^3 + 3x^2 + 5$?
 - A. 3
 - B. 5
 - C. 12
 - D. 17
32. What is the sum of the coefficients in the expression $a - 3b - c + 2d$?
 - A. -3
 - B. -1
 - C. 2
 - D. 3

Answers and explanations begin on page 673.

Lesson 9: Simplify Polynomials

Key Ideas

- Like terms can be combined; unlike terms cannot.
- Like terms are either numbers—positive or negative—or variables with the same degree of exponents.
- Polynomials with unlike terms cannot be simplified.

Combine Like Terms

When working with polynomials, you must work to combine like terms. **Like terms** have variables that are the same or are terms that have the same variable raised to the same power, or exponent. For example, $8x^3$ and $4x^3$ are like terms, and $5y$ and $3y$ are also like terms.

Unlike terms have variables that are different or have the same variable raised to different powers. For example, $3a$ and $5b$ are unlike terms. The terms $9x$ and $7x^2$ are also unlike terms because the variables are raised to different powers.

You can only combine like terms when simplifying polynomials. Unlike terms cannot be combined. For example, in the polynomial $7x^2 + 4x^3$, you cannot combine the terms because x is raised to different powers in each term. However, in the polynomial $7x^2 + 4x^2$, you can combine terms that all have the same variable and the same exponent: $7x^2 + 4x^2 = 11x^2$.

Example 1: Combine like terms in the polynomial $4y^3 + 2x^3 - 7y^3$.

1. First, identify and group the like terms. $4y^3 - 7y^3 + 2x^3$

The two y^3 terms are alike; the x^3 term cannot be combined with them.

2. Combine the like terms. $-3y^3 + 2x^3$

Simplify Polynomials

Some polynomials may contain both like and unlike terms. The following example shows how like and unlike terms are handled when you have to simplify a polynomial. Start by combining like terms with the largest exponent, and proceed to the terms with the smaller exponents.

Example 2: Simplify the polynomial $7x^4 - 13x^2 + x^4 + 2x^2 + 5x + 3 + x$.

1. Combine like terms with the largest degree, which is 4. Add:
 $7x^4 + x^4 = 8x^4$.
2. You now have $8x^4 - 13x^2 + 2x^2 + 5x + 3 + x$. Combine like terms from the exponent with the next largest degree, which is 2. Add:
 $-13x^2 + 2x^2 = -11x^2$.
3. You now have $8x^4 - 11x^2 + 5x + 3 + x$. Finally, combine like terms with the next largest degree, which is 1. Add: $5x + x = 6x$.
4. You now have $8x^4 - 11x^2 + 6x + 3$.

This is as much as you can simplify this polynomial by combining like terms. The simplified polynomial has no more like terms and must remain as is.

GED® TEST TIP

When working with polynomials, treat the minus sign before a coefficient or constant as an indicator of a negative number. This will help you to combine like terms using the rules of signed numbers that you learned on pages 316–318.

Practice 9

A. For each pair of terms, indicate if they are like terms or unlike terms.

1. $4x^3, x^3$ _____ terms
2. $3x, x$ _____ terms
3. b, b^2 _____ terms
4. $-2x, 7y$ _____ terms
5. $-2x, 7x$ _____ terms
6. $4a, 4$ _____ terms
7. g^2, g^2hi _____ terms
8. $2x^2y, 8x^2y$ _____ terms
9. $-5m, -5m^2$ _____ terms
10. x^2y, xy^2 _____ terms
11. $ab, 8ab$ _____ terms
12. $3y, 3y^2$ _____ terms
13. $14g, \frac{1}{3}g$ _____ terms
14. $12x^2, x^2$ _____ terms
15. a, ab^2 _____ terms
16. $15x^2, -x^2$ _____ terms
17. $10y, 11yz$ _____ terms
18. $x^2, \frac{x^2}{2}$ _____ terms
19. $g, -g^2$ _____ terms
20. $\frac{x}{8}, x^2$ _____ terms

B. Simplify each expression by combining like terms.

21. $3x^2y + 4x^2y$
22. $3b + b$
23. $a - 7a + 3$
24. $14ab + ab^2 + 2ab + 3$
25. $x^2 - 3x^2 + 7$
26. $ab + 2ab + ab$
27. $7g + 7gh + 7g + 7gh$
28. $g^2 + h^2 - 4g^2h^2 + g^2 + h^2 - 4$
29. $9y + y - y^2 - y$
30. $x^2 - 8x^2 + y - 3$
31. $11y + 11y - 7$
32. $9x^2y - 3x^2y + 4y^2 - 21 + y^2 - 2$
33. $8x^2 - 4x + 7 + 4x^3 - x^2 + 7x - 2$
34. $-3x^2 + 6x - 2x^2 - 10x + 5$
35. $9x^2 - 3x^3 + x - 2x^2$

C. Choose the one best answer to each question.

36. What is the simplified form of the expression $3a^2b + 4ab + 3a^2b + 5ab$?
 A. $6a^3b^2 + 8a^3b^2$
 B. $6a^2b + 9ab$
 C. $6a^2b + 8ab$
 D. $15a^6 + b^4$
37. When simplified, how many terms are in the polynomial $j^2 + k^3 - 2j^3 + 5k^3 - 2j^2$?
 A. 2
 B. 3
 C. 4
 D. 5

Answers and explanations begin on page 673.

Lesson 10: Add and Subtract Polynomials

Key Ideas

- Monomials, binomials, or trinomials can be added or subtracted.
- In some cases, a polynomial needs to be simplified before it can be added or subtracted.
- When subtracting polynomials, distribute the second polynomial, changing the signs for coefficients, and then combine all the like terms.

GED® TEST TIP

When you subtract, you are simply reversing the signs within the second polynomial. Be careful to make all negative signs positive. Study Examples 3 and 4 carefully to avoid common errors.

Add Polynomials with One Variable

On the *Mathematical Reasoning Test*, you may be required to add polynomials that have one variable. You can add polynomials when they have like terms, including variables and exponents. When you see polynomials within parentheses linked by an addition sign, add them by combining the like terms across the polynomials.

Example 1: Add $(3x^2 + x + 4) + (2x^2 + 2x - 16)$.

1. To add these polynomials, simply combine like terms. Add:

$$3x^2 + 2x^2 = 5x^2.$$

2. Add: $x + 2x = 3x$.

3. Add: $4 + (-16) = -12$.

4. Combine the results into one polynomial: $5x^2 + 3x - 12$.

Sometimes, an expression needs to be simplified before you add and combine like terms.

Example 2: Add $(5x^2 + 8x - 4) + (2x^2 - 6x + 14x)$.

1. Simplify within the second parentheses. Add: $-6x + 14x = 8x$.

2. Combine like terms across the polynomials. Add: $5x^2 + 2x^2 = 7x^2$.

3. Use the simplified expression from step 1. Add: $8x + 8x = 16x$.

4. Combine the results into one polynomial: $7x^2 + 16x - 4$.

Subtract Polynomials with One Variable

The only difference between adding and subtracting polynomials is the minus sign between the parentheses. You must distribute the negative sign across the second polynomial. To distribute the negative sign across the second polynomial, simply reverse the signs of each term. Then drop the parentheses and combine like terms across both polynomials.

Example 3: Subtract $(3x^2 + x + 4) - (7x^2 + 2x - 16)$.

1. Distribute the negative sign to everything in the second set of parentheses by reversing the sign of each term: $-7x^2 - 2x + 16$.
2. Drop the parentheses from the entire expression and combine like terms:

$$3x^2 + x + 4 - 7x^2 - 2x + 16$$

$$= -4x^2 + x + 4 - 2x + 16$$

$$= -4x^2 - x + 4 + 16$$

$$= -4x^2 - x + 20$$

Note that subtracting a negative constant or coefficient is the same as adding a positive. In the next example, watch what happens to the negative coefficient inside the second set of parentheses when the sign is distributed.

Example 4: Subtract $(x^2 + 2x + 3) - (4x^2 - x + 6)$.

1. Distribute the negative sign to everything in the second set of parentheses by reversing the sign of each term:

$$-4x^2 + x - 6$$

2. Drop the parentheses and combine like terms:

$$x^2 + 2x + 3 - 4x^2 + x - 6$$

$$x^2 - 4x^2 + 2x + x + 3 - 6$$

$$-3x^2 + 3x - 3$$

Practice 10.1

A. Add the following polynomials.

1. $(3x + 4) + (2x + 2)$

2. $(17y - y + 3) + (4y + 3y + 3)$

3. $(5x^2 - 3x - 4) + (3x^2 - 2x + 6)$

4. $(-a^2 + 2a) + (16a^2 + 6a)$

5. $(9x^2 - 3x - 2) + (2x^2 + 5x + 5)$

6. $(6a + 6) + (-5a - 5)$

7. $(-8g^2 + 7g + 6) + (8g^2 - 7g - 5)$

8. $(2x^2 + 5x) + (2x^2 + 4x + 7x - 9)$

9. $(13y + 4y + 4) + (7y - 7)$

10. $(-a^2 - a^2 - a - 4) + (-a^2 - a^2 - a - 5)$

B. Subtract the following polynomials.

11. $(3y - 4) - (2y - 2)$

12. $(x + 16) - (4x + 3)$

13. $(2a + 1) - (-a - 1)$

14. $(5x^2 + 2x + 4) - (2x^2 + x + 2)$

15. $(7y + 5y + 5) - (2y - 2)$

16. $(9x^2 + 4x + 5x + 4) - (7x^2 + 6x - 9)$

17. $(-g + g - 1) - (-g - g - 2)$

18. $(17a^2 - 4a - 4) - (16a^2 + 6a - 6)$

19. $(7b^2 + b - 8) - (7b^2 + b + 8)$

20. $(21x^2 + 3x^2 - 2x - 4 - 1) - (3x^2 + x^2 + x + 2x - 5 - 1)$

C. Choose the one best answer to each question.

21. What is the sum of the polynomials

$$(2xy + 3xy^2 - 4x^2y) + (5x^2y - 3xy^2 + 2xy)?$$

A. $x^2y + 4xy$

B. $4xy + 3x + y$

C. $7x - 2y$

D. $7x^2y - 2xy$

22. Which of the following equals

$$(5x^2 - 2x + 1) - (3x^2 - 3x - 2)?$$

A. $x - 5x - 1$

B. $2x^2 - x + 3$

C. $2x^2 + x + 3$

D. $2x^2 - 5x + 3$

Answers and explanations begin on page 673.

Simplifying Polynomials with More Than One Variable

Some polynomials contain terms with different variables—for example, the binomial $2x + 3y$ contains an x term and a y term. You add, subtract, and simplify polynomials with more than one variable just as you do polynomials with just a single variable. However, you have to be careful to add and subtract only like terms. Remember that in all polynomials, like terms have the *same variable* and the *same exponent*. For example, in the polynomial $x^2 + 2x + y^2 + 3x + 2y$, the only like terms are the two x^1 terms. You can combine the $2x$ and the $3x$, but you cannot combine x^2 with any other term, nor can you combine the y^2 or the $2y$ with any other term. The most simplified version of this polynomial is $x^2 + 5x + y^2 + 2y$.

A term that consists of two or more variables multiplied together cannot be combined with terms that contain only one of those variables. For example, you cannot combine $2xy + 7x$. However, you can combine $2xy + 7xy$ to get $9xy$.

Example 1: Simplify $6xy^4 - 20x^2 - 3 + xy^4 + 12x^2 - 30$.

Start by grouping the like terms together. Then add and subtract as appropriate.	$\begin{aligned} &6xy^4 - 20x^2 - 3 + xy^4 + 12x^2 - 30 \\ &= 6xy^4 + xy^4 - 20x^2 + 12x^2 - 3 - 30 \\ &= 7xy^4 - 8x^2 - 33 \end{aligned}$
--	--

Adding Polynomials with More Than One Variable

Adding polynomials that contain more than one variable requires the same process as adding polynomials that have only one variable. Just remove the brackets and combine like terms across both polynomials. Remember that only like terms can be combined.

Example 2: Add $(3x^2 + 6x + 3xy) + (5x^2 - 6xy + 14y)$.

$$\begin{aligned} &(3x^2 + 6x + 3xy) + (5x^2 - 6xy + 14y) \\ &= 3x^2 + 5x^2 + 6x + 3xy - 6xy + 14y \\ &= 8x^2 + 6x - 3xy + 14y \end{aligned}$$

Subtracting Polynomials with More Than One Variable

Subtracting polynomials that contain more than one variable is exactly the same as subtracting polynomials that contain only a single variable. Remember to distribute the minus sign across the second polynomial when you remove the parentheses. Also, remember that only like terms can be combined.

Example 3: Subtract $(3xy^2 + x + 2xy + 4) - (7xy^2 + 2xy - 16)$.

$$\begin{aligned} &(3xy^2 + x + 2xy + 4) - (7xy^2 + 2xy - 16) \\ &= 3xy^2 + x + 2xy + 4 - 7xy^2 - 2xy + 16 \\ &= -4xy^2 + x + 20 \end{aligned}$$

Notice that there is no xy term in the final simplified expression because $2xy - 2xy = 0$.

Practice 10.2

A. Simplify the following polynomials.

1. $a^2 + 2a + b^2 + 3a + 3b^2$
2. $x + 3xy + y + 16 - 5x - 4xy$
3. $yz^2 - z^2 + 5 + 3yz - 3z^2 + 4yz$
4. $ab - b + c + 17 + 6ab - 34$
5. $a^4 + c^2 + 3ac - 5a^4 - 2c^2 + 5$
6. $19xyz + x - y + 3xyz + 6 + y$
7. $ef + ef^2 - 10 + 6ef + f$
8. $p + pq - q - p + 3pq - 3 - q + 7$

B. Add the following polynomials.

9. $(ab + b - c + 6) + (3ab - b + c - 6)$
10. $(x^2 + xy - y - 3) + (-x^2 - xy + y + 3)$
11. $(p^6 + p^5 + pq - p^2) + (2p^6 - p^5 + 3pq - q^2)$
12. $(x + xy + 7) + (y - xy + 8)$
13. $(9a + bc + c^2) + (-9a - b + bc - c^2)$
14. $(4xyz + 4x - 4y) + (6xyz - 4x - 4y + 16)$
15. $(5a + 5b + 4c - 6) + (a - b - 4c + d + 6)$
16. $(pq - q + 15) + (6pq - 15)$

C. Subtract the following polynomials.

17. $(ab + a + b + 6) - (8ab + a - b - 6)$
18. $(xy + x + y) - (xy + x + y)$
19. $(p^2 + q^2 + 14) - (3p^2 - 4q^2 + 7)$
20. $(x^4 + yz + z^4) - (yz + z^5)$
21. $(bc^2 + bc + c - 7) - (2bc - c - 7)$
22. $(xy + xz - xyz) - (4xy + xyz)$
23. $(c^3d^7 + c^2 - d + 35) - (2c^2 - 3d + 35)$
24. $(ef^2g - fg + 16) - (ef^2 + fg + 8)$

D. Choose the one best answer to each question.

25. Which of the following equals $(x + y + z) - (-2x + 2y + z)$?
A. $-x + 3y + 2z$
B. $-2x - y + 2z$
C. $3x - 2y + 2z$
D. $3x - y$
26. Which of the following equals $(2g^2 + 3h - 5k) + (g^2 + 2h + 3k)$?
A. $g^2 + h - 8k$
B. $3g^2 + 5h - 2k$
C. $2g^2 + 6h^2 - 15k^2$
D. $2g^4 + 5h - 2k$

Answers and explanations begin on page 674.

Lesson 11: Multiply Polynomials

Key Ideas

- There are a number of ways to multiply polynomials.
- The basic concepts in all involve learning procedures for multiplying constants, variables, and exponents.
- A common method for multiplying two binomials is called FOIL for short.

GED® TEST TIP

Always finish the FOIL method by combining like terms after you have multiplied.

Multiply a Monomial by a Monomial

To multiply two monomials, multiply each component separately. Start by multiplying the coefficients (or numbers). If there is no coefficient given, assume that the coefficient is 1 (for example, $xy = 1xy$). Then, multiply each variable separately by adding the exponents of like variables.

Remember that a variable with no exponent is a variable raised to the first power (for example, $x = x^1$). So $x(x^3) = x^1(x^3) = x^4$.

Example 1: Multiply $(2ab)(4b)$.

$$2 \times 4 \times a \times b^1 \times b^1 = 8ab^2$$

Multiply a Monomial by a Binomial

To multiply a monomial by a binomial, multiply the monomial by the first term in the binomial, then multiply the monomial by the second term in the binomial. Finally, add the resulting terms.

Example 2: Multiply $-2z^2(z + 3yz)$.

1. Distribute the negative sign:

$$-2z^2(z + 3yz) = 2z^2(-z - 3yz).$$

2. Multiply the monomial by each term in the binomial:

$$2z^2(-z - 3yz) = [2 \times (-1) \times z^2 \times z^1] + [2 \times (-3) \times z^2 \times y^1 z^1].$$

3. Add: $-2z^3 + (-6yz^3) = -2z^3 - 6yz^3$.

Multiply Two Binomials

To multiply two binomials, multiply the first terms, then the outer terms, then the inner terms, and, finally, the last terms. Add the results and combine like terms. You can remember this process by the acronym FOIL, which stands for First, Outer, Inner, Last.

Example 3: Multiply $(2g^2 + 9)(3g^2 - 4)$.

1. Multiply using FOIL:

$$(2g^2 + 9)(3g^2 - 4) = \overbrace{(2g^2)(3g^2)} + \overbrace{(2g^2)(-4)} + \overbrace{(9)(3g^2)} + \overbrace{(9)(-4)}.$$

2. Simplify each product: $6g^4 - 8g^2 + 27g^2 - 36$.

3. Combine like terms:

$$6g^4 - 8g^2 + 27g^2 - 36 = 6g^4 + 19g^2 - 36.$$

Practice 11

A. Multiply the monomials.

1. $(6x)(5x)$
2. $(2xy)(3y)$
3. $(7abc)(4bc)$
4. $(12y)(z)$
5. $(a)(9bc)$
6. $(5xyz)(2xy^2z^4)$
7. $(4ab)(bc^2)$
8. $(17f^2gh^3)(2fh^4)$

B. Multiply the binomials by the monomials.

9. $3z^2(6xy + 4z)$
10. $6x(7x - 6z)$
11. $-5ab(3b + 11c)$
12. $-3f^3(6h - 8fgh^2)$
13. $10z(7x^7 - 5z)$
14. $-z(-z - 6xy)$
15. $8b(9ab + 8a)$
16. $-9(-2x^4 + 3xy^2)$

C. Multiply the binomials by using FOIL. Remember to complete by combining like terms wherever possible.

17. $(x + 5)(x - 6)$
18. $(x + y)(x + y)$
19. $(z + 9)(z - 9)$
20. $(yz^2 + x)(yz^2 - 3x)$
21. $(3x + 3)(3x + 5)$
22. $(x + y)(x - y)$
23. $(y^2 - 6)(y^4 + 10)$
24. $(ab + 3)(ab - 4)$

D. Choose the one best answer to each question.

25. Which of the following expressions is equal to $(4a^3b^2)(3a^2c)$?
 - A. $12a^5b^2c$
 - B. $7a^5b^2c$
 - C. $12a^6bc$
 - D. $7a^6b^2c$
26. Martha has $2pc^2$ board games, and John has $6p^2c$ board games. Which of the following expressions represents the product of the number of board games that Martha and John have?
 - A. $8p^3c^3$
 - B. $8p^2c^2$
 - C. $12p^2c^2$
 - D. $12p^3c^3$
27. Which of the following expressions is equal to $(4ab + 2)(3ab - 7)$?
 - A. $7a^2b^2 + 22ab + 5$
 - B. $12a^2b^2 - 22ab - 14$
 - C. $12a^2b^2 + 22ab + 14$
 - D. $7a^2b^2 - 22ab - 5$

Answers and explanations begin on page 674.

Lesson 12: Divide Polynomials

Key Ideas

- You can divide a polynomial by numbers, expressions, or other polynomials.
- The key to dividing a polynomial is to isolate a common factor that divides into the numerator and denominator.
- Common factors can be numbers, variables, or a combination of the two such as addition, subtraction, or multiplication expressions.

Divide Polynomials by a Number

The *Mathematical Reasoning Test* will sometimes ask you to simplify a polynomial that is written as a fraction. To simplify, divide the numerator and denominator by a common term. A common term can be the following:

- a number
- a variable with a number as a coefficient
- a more complicated expression

You may need to factor the common term out of the numerator or denominator before you can do the division.

Example 1: Simplify $\frac{3x+6}{3}$.

$$\frac{3x+6}{3} = \frac{3(x+2)}{3} = \frac{\cancel{3}(x+2)}{\cancel{3}} = x+2$$

Divide Polynomials by an Expression

Dividing a polynomial by an expression is really no different from dividing by a number. You need a common term in the numerator and denominator, and you may have to factor out that common term (including any variables) before you can do the division.

Example 2: Simplify $\frac{25x^3-45x^2}{5x}$.

$$\frac{25x^3-45x^2}{5x} = \frac{5x(5x^2-9x)}{5x} = \frac{\cancel{5x}(5x^2-9x)}{\cancel{5x}} = 5x^2-9x$$

When the numerator has two added or subtracted expressions that both include the common term, it may be easier to split the polynomial into two separate fractions and simplify them separately. Doing it this way will help you keep straight exactly which terms you can cancel.

Example 3: Simplify $\frac{x(x+7)-3(x+7)}{x+7}$.

$$\begin{aligned}\frac{x(x+7)-3(x+7)}{x+7} &= \frac{x(x+7)}{x+7} - \frac{3(x+7)}{x+7} \\ &= \frac{\cancel{x(x+7)}}{\cancel{x+7}} - \frac{\cancel{3(x+7)}}{\cancel{x+7}} = x-3\end{aligned}$$

Undefined Expressions

When the denominator of an expression equals zero, the expression is **undefined**.

Example 4: What is the value of $\frac{x^2+2xy+y^2}{y-x}$ when $x=y$?

Because $x=y$, $y-x=y-y$. Because $y-y=0$, this expression is undefined.

GED® TEST TIP

Remember that you will be provided with a wipe-off board during the Mathematical Reasoning Test. You will be able to use that board to carefully work through the steps needed to solve these types of problems.

Practice 12

A. To simplify, divide each polynomial by a number. (You may need to factor the numerator and/or denominator to find the common term.)

1. $\frac{2y + 30}{2}$

2. $\frac{7x + 21}{7}$

3. $\frac{4x + 20}{4}$

4. $\frac{3a + 3b}{3}$

5. $\frac{11x^2 + 22x}{11}$

6. $\frac{26x^2 + 39x + 13}{13}$

7. $\frac{5x + 10y}{5}$

8. $\frac{21a + 14b}{7}$

9. $\frac{48x + 32y}{32}$

10. $\frac{9a - 15b}{12}$

11. $\frac{6b + 24c}{18}$

12. $\frac{25a + 5b + 15c}{10}$

B. Divide each polynomial. Decide whether it is easier for you to divide the original numerator by the common term (as in Example 2) or whether to split the numerator first (as in Example 3).

13. $\frac{18x^2 + 6x}{3x}$

14. $\frac{10x^2 + 6x}{2x}$

15. $\frac{40y^2 + 10y}{5y}$

16. $\frac{42xy + 49x}{7x}$

17. $\frac{38xy + 38x}{19x}$

18. $\frac{3x + 18}{x + 6}$

19. $\frac{x(x + 4) - 6(x + 4)}{x + 4}$

20. $\frac{y(y + 3) + 7(y + 3)}{y + 3}$

21. $\frac{z(z + 2) - 5(z + 2)}{z + 2}$

22. $\frac{22x^2 + 66x}{11x}$

23. $\frac{x(y - 2) - 6(y - 2)}{y - 2}$

24. $\frac{a(b + 3) + 3(b + 3)}{b + 3}$

C. Choose the one best answer to each question.

25. Which of the following is equal

to $\frac{21a + 14b}{7}$?

- A. $5a + 5b$
- B. $3a + 14b$
- C. $3a + 2b$
- D. $21a + 2b$

27. Which of the following expressions is

equal to $\frac{2z^2 - 12z + 2(z + 4) - 8}{2z}$?

- A. $z - 5$
- B. $z(z + 2) - 5$
- C. $z(z + 2) - 5(z + 2)$
- D. $z - 3$

26. James has $27x$ apples, Rachel has $51y$ oranges, and Glen has $60z$ peaches. Which of the following expressions represents the average number of pieces of fruit that James, Rachel, and Glen have?

- A. $9x + 51y + 60z$
- B. $46x + 46y + 46z$
- C. $9x + 17y + 60z$
- D. $9x + 17y + 20z$

28. For what values of a is the expression

$\frac{a(4a - 16a)}{a \times |a| + a^2}$ undefined?

- A. $a = 4$
- B. $a \leq 0$
- C. $a = 16$
- D. $a \geq 0$

Answers and explanations begin on page 674.

Algebra Basics, Expressions, and Polynomials

Practice Questions

1. Which of the following expressions is equal to $6 - 4(x + 3)$?

A. $4x + 3$
 B. $4x - 9$
 C. $-4x + 9$
 D. $-4x - 6$

2. If Kris makes d dollars and Heidi makes 75 dollars less than 3 times Kris's wage, what does Heidi make in terms of d ?

A. $d + 75$
 B. $d - 225$
 C. $3d - 75$
 D. $3d - 225$

3. What is the product of 700 and 180,000 written in scientific notation?

A. 12.6×10^9
 B. 1.26×10^8
 C. 1.26×10^9
 D. 1.26×10^{10}

4. If there are $4x$ identical schools in a region and each school has $3y$ classrooms each with $7x$ desks, how many desks are there in the region?

A. $11x + 3y$
 B. $28x + 3y$
 C. $84xy^2$
 D. $84x^2y$

5. Which of the following is equivalent to $x^2 - 25$?

A. $x(x - 25)$
 B. $(x - 5)^2$
 C. $(x + 5)(x - 5)$
 D. $(x - 25)^2$

6. If Tom has $9x$ baseball cards, Adam has 13 more than Tom, and Dave has $2y$ baseball cards, together they would have how many cards?

A. $9x + 2y + 13$
 B. $18xy + 13$
 C. $18xy + 2y$
 D. $18x + 2y + 13$

Questions 7 and 8 refer to the figure below.

The figure below is a multiplication box. Each place represents the horizontal number multiplied by the vertical number; for example, the 9 in the lower right corner equals 3 multiplied by 3.

	5	y	3
1	5		3
c	20	a	
3	b	d	9

7. Which of the following is equal to ab ?

A. $5y$
 B. $60y$
 C. $100y$
 D. $500y$

8. Which of the following is equal to $a - d$?

A. b
 B. c
 C. x
 D. y

9. Multiply $7c^2(a^2 + 5b + 7c^2)$.
- $7a^2c^2 + 35bc^2 + 49c^4$
 - $7(ac)^2 + 35(bc)^2 + 49c^4$
 - $7a^2 + 35b + 49c^4$
 - $a^2 + 5b + 14c^2$
10. A Little League pie sale fundraiser generated \$900 in revenue. If there were $6x$ pies sold, how much did each pie cost?
- \$10
 - \$15
 - $\$ \frac{x}{150}$
 - $\$ \frac{150}{x}$
11. Simplify the expression
- $$\frac{(a-4)^2(6b)}{2(3b)(a+4)(a-4)}$$
- $\frac{a-4}{a+4}$
 - $\frac{a+4}{a-4}$
 - $\frac{a-2}{2}$
 - 1
12. Emilie buys x ounces of chicken for each of her guests at a dinner party. There are $8y$ people coming, and chicken is \$ z a pound. How much will the total amount of chicken cost in dollars? (*Hint:* There are 16 ounces in one pound.)
- $8xyz$
 - $16xyz$
 - $\frac{xyz}{2}$
 - $\frac{xyz}{8}$
13. What is the value of the expression $6(x - y) - 8x$ when $x = -2$ and $y = 5$?
- 58
 - 26
 - 2
 - 34
14. Which of the following is equal to $x^2 + 6z$ multiplied by $2y - 4z$ if $x = 2$, $y = 4$, and $z = 3$?
- 88
 - 72
 - 0
 - 104
15. What is the value of the following expression?
- $$\frac{4^3 - [3(12 + 2^2)]}{6 + 5(4) - 15}$$
- Write your answer on the line below.
-
16. If every member of a team is paid $5p$ dollars for his or her participation, and Team Alpha has $6k$ members, Team Beta has $4p$ members, and Team Delta has $9h$ members, which of the following represents the total amount paid, in dollars, to the members of all three teams?
- $1080hkp$
 - $1080hkp^2$
 - $120p^2k + 45ph$
 - $30kp + 20p^2 + 45hp$

17. Which of the following is equal to $7x - [(5y)2x - 9y + 8x]$?

- A. $x - 9y + 10xy$
- B. $-3x - 4y$
- C. $-x + 9y - 10xy$
- D. $15x - 9y + 10xy$

18. If Stephanie has $5c$ CDs for every DVD Alyshia has, and Jeff has $4d$ DVDs, which is 15 more than Alyshia has, how many CDs does Stephanie have?

- A. $20cd - 75c$
- B. $20cd + 15$
- C. $80cd - 15$
- D. $80cd - 75c$

19. Divide $\frac{56x^4 + 49x^3y^3 - 84x^3 - 7x^2}{7x^2}$.

- A. $8x^2 + 7xy^3 - 12x - 1$
- B. $8x^4 + 7x^3y^3 - 12x^3 - x^2$
- C. $8x^4 + 7y^3 - 5x^3 - 1$
- D. $49x^2 + 42xy^3 - 77x$

20. A charity with a yearly budget of $80d$ gets all of its money from either its yearly ball or from private donations. If the charity raised $64d$ from the silent auction at the ball and $4a$ from ticket sales at the ball, how much will need to be raised in private donations to meet the budget?

- A. $16d$
- B. $16d - 4a$
- C. $16d + 4a$
- D. $144d - 4a$

21. Divide: $\frac{x^4 + 4x^2y^4 - 8x - y^2}{2y^2}$.

- A. $\frac{x^4}{2y^2} + 2x^2y^2 - \frac{4x}{y^2} - \frac{1}{2}$
- B. $\frac{x^2}{2y} + \frac{2x^2y^2 - 4x}{y^2 - \frac{1}{2}}$
- C. $\frac{2x^4y^2 + 8x^2y^2 - 16x}{y^2 + 2y^4}$
- D. $\frac{x^4 - 4x}{2y^2 + 2x^2y^2 - \frac{1}{2}}$

22. If $12k$ privates, $8f$ lieutenants, and $6r$ captains are divided into $2r$ equal squads, which of the following would express the number of members of one squad?

- A. $6k + 4f + 3r$
- B. $\frac{6k + 4f}{r} + 3$
- C. $12k + 8f + 4r$
- D. $12k + 8f + 3r$

23. If $x \neq 0$, $q = 2$, and $r = 4$, which of the following would equal $\frac{r^2x^2 + q^3x}{(q^2x)}$?

- A. $4x + 2$
- B. $4x^2 + 2$
- C. $16x^2 + 8x - 2$
- D. $16x^2 + 21$

24. Simplify:

$$\frac{44x^6 + 55x^3y^4 + 88x^2 - 11y^2}{11x^2} - 6x^4 - 8$$

- A. $4x^4 + 5xy^2 + 8x - 1$
- B. $2x^4 + 5xy^3 - y^2 - 8$
- C. $-2x^4 + 5xy^4 - \frac{y^2}{x^2}$
- D. $10x^4 + 5y^4 - \frac{y^2}{x^2} - 8$

25. Simplify:

$$(6x^2 + 5xy^2)(y) - (4xy - 9x)(4xy).$$

- A. $5xy^3 - 16x^2y^2 + 42x^2y$
- B. $5xy^3 - 16x^2y^2 - 30x^2y$
- C. $6x^2y - 11x^2y^2 - 36xy^2$
- D. $6x^2y + 5x^2y^2 - 36xy^2 + 16xy$

26. There are $4x$ workers, including managers, in an office. If the salaries of the 3 managers are not included, the average salary of each employee is $5x - 6$ dollars. What do the nonmanagerial employees earn in total?

- A. $20x^2 - 24x$
- B. $20x^2 - 24x - 3$
- C. $20x^2 - 9x + 18$
- D. $20x^2 - 39x + 18$

27. What is the value of $\frac{45xy + 63y^2}{9y}$ if $x = 6$ and $y = 3$?

- A. 12
- B. 51
- C. 108
- D. 153

28. Simplify:

$$\frac{3x(x^2 + 3) - 2x(x^2 + 3) - (x^2 + 3)}{x^2 + 3}$$

- A. $x^3 - 1$
- B. $x^3 - 3x - 1$
- C. $3x - 1$
- D. $x - 1$

29. If the $3h$ people in one country and the $4k$ people in the neighboring country each consume $h - 17$ pounds of rice per year on average, how many pounds of rice would be consumed by both countries in one year?

- A. $4h + 4k - 17$
- B. $4h^2 + 4k - 68k$
- C. $12hk + h - 17$
- D. $3h^2 + 4hk - 51h - 68k$

30. Simplify:

$$\frac{4x^2(x + y) - 2(x + y)^2(x - y) - xy(x + y)}{x + y}$$

- A. $2x^2 + 2y^2 - xy$
- B. $4x^2 + xy - 2x - 2y$
- C. $4x^2 - xy - 2x - 2y$
- D. $3x^2 - 2xy^2 - xy$

Questions 31 and 32 refer to the figure below.

The figure below is a multiplication box. Each place represents the horizontal number multiplied by the vertical number; for example, the 20 in the lower right corner equals 5 multiplied by 4.

	a^2	ab	5
a			ab
c	$8b$		
4		40	20

31. Which of the following is equal to a ?

- A. 2
- B. 4
- C. 5
- D. 10

32. Which of the following is equal to a^2c ?

- A. 13
- B. 40
- C. 42
- D. 60

Answers and explanations begin on page 674.

Equations, Inequalities, and Functions

Lesson 1: Equations

Writing and Solving One-Step Equations

An **equation** is a mathematical statement that two expressions are equal.

Examples: $3 + 5 = 4 \times 2$ $10 - 1 = 3^2$ $5(3 + 4) = 35$

An equation can contain one or more variables. Solving an equation means finding a value for the variable that will make the equation true.

Examples: $4 + x = 11$ $3x = 24$ $x - 5 = -2$
 $x = 7$ $x = 8$ $x = 3$

The basic strategy in solving an equation is to isolate the variable on one side of the equation. You can do this by performing **inverse**, or **opposite**, operations. However, you must always follow one basic rule: whatever you do to one side of the equation, you must also do to the other side.

Example 1: Solve $x - 23 = 45$.

On the left side of the equation, the number 23 is subtracted from x . The inverse of subtraction is addition. Add 23 to both sides of the equation.

$$\begin{aligned} x - 23 &= 45 \\ x - 23 + 23 &= 45 + 23 \\ x &= 68 \end{aligned}$$

To check your work, replace the variable with your solution and simplify:
 When $x = 68$, the equation is true.

$$\begin{aligned} \text{Check: } x - 23 &= 45 \\ 68 - 23 &= 45 \\ 45 &= 45 \end{aligned}$$

The following examples use the inverse operations of multiplication and division.

Example 2: Solve $\frac{x}{2} = 17$.

The variable x is divided by 2. Since multiplication is the inverse of division, you must multiply each side of the equation by 2.

When $x = 34$, the equation is true.

$$\begin{aligned} \frac{x}{2} &= 17 \\ 2\left(\frac{x}{2}\right) &= 2(17) \\ x &= 34 \\ \text{Check: } \frac{34}{2} &= 17 \\ 17 &= 17 \\ 5x &= 75 \end{aligned}$$

Example 3: Solve $5x = 75$.

Since the variable x is multiplied by 5, divide both sides of the equation by 5.

$$\begin{aligned} \frac{5x}{5} &= \frac{75}{5} \\ x &= 15 \end{aligned}$$

When $x = 15$, the equation is true.

$$\begin{aligned} \text{Check: } 5(15) &= 75 \\ 75 &= 75 \end{aligned}$$

Key Ideas

- An equation states that two expressions are equal.
- To solve an equation, isolate the variable by applying inverse operations to both sides of the equation.
- Check your work by substituting your solution for the variable in the original equation and simplifying.

GED® TEST TIP

If you are unsure about a multiple-choice algebra problem, substitute each of the answer choices for the variable in the equation. You may be able to do this quickly using your calculator.

Practice 1.1

A. Solve for the variable in each equation.

1. $7x = 63$
2. $23 + m = 51$
3. $-13 = y - 12$
4. $\frac{x}{4} = -16$
5. $5a = 625$
6. $y - 17 = -30$
7. $x + 6 = 33$
8. $4c = 28$
9. $\frac{12}{x} = -3$
10. $26 = b + 33$
11. $93 = 3x$
12. $s + 16 = 8$
13. $36 = \frac{x}{3}$
14. $t + 14 = 53$
15. $\frac{x}{6} = 8$
16. $16y = -48$
17. $r - 35 = 75$
18. $24 = \frac{120}{x}$
19. $5y = -45$
20. $d + 45 = 20$
21. $16 = 4x$
22. $-4x = 24$
23. $19 = h - 7$
24. $\frac{x}{11} = 6$
25. $m + 24 = 14$
26. $5y = 45$
27. $14 - w = 42$
28. $18 = \frac{y}{4}$

B. Choose the one best answer to each question.

Questions 29 and 30 refer to the following table.

April Time Sheet Summary
Hours Worked per Week

Week	1	2	3	4
Kayla Sax	36	40	40	
Erin Grady		24	28	38

29. Kayla and Erin worked a total of 77 hours during Week 1. Let x = Erin's hours for Week 1. Which of the following equations could be used to solve for Erin's hours during Week 1?
 - A. $x - 36 = 77$
 - B. $x + 77 = 36$
 - C. $x + 36 = 77$
 - D. $x - 77 = 36$
30. Erin worked twice as many hours as Kayla did during Week 4. Let y = Kayla's hours for Week 4. Which of the following equations could be used to solve for Kayla's hours during Week 4?
 - A. $\frac{y}{2} = 38$
 - B. $38y = 2$
 - C. $2y = 38$
 - D. $\frac{1}{2y} = 38$
31. A number divided by 4 is 32. What is the number?
 - A. 8
 - B. 28
 - C. 128
 - D. 512
32. The solution $x = -5$ makes which of the following equations true?
 - A. $14 - x = 9$
 - B. $\frac{x}{5} = 1$
 - C. $x + 3 = 8$
 - D. $12x = -60$
33. Mike had \$572.18 in his checking account. After writing a check, he had \$434.68. Which of the following equations could be used to find the amount of the check (c)?
 - A. $\$572.18 + c = \434.68
 - B. $\$572.18 - c = \434.68
 - C. $\$572.18c = \434.68
 - D. $\frac{\$572.18}{c} = \434.68

Answers and explanations begin on page 676.

Solving Multi-Step Equations

Most equations require more than one operation in order to find a solution. Follow these basic steps:

- Simplify by combining like terms.
- Perform addition and subtraction steps.
- Perform multiplication and division steps.

Example 1: Solve $6x + 5 - 2x = 25$.

1. Combine like terms ($6x - 2x = 4x$).

$$6x + 5 - 2x = 25$$

$$4x + 5 = 25$$

2. Subtract 5 from both sides.

$$4x + 5 - 5 = 25 - 5$$

$$4x = 20$$

3. Divide both sides by 4.

$$\frac{4x}{4} = \frac{20}{4}$$

$$x = 5$$

4. Check by substituting the solution for x in the original equation.

$$6(5) + 5 - 2(5) = 25$$

$$30 + 5 - 10 = 25$$

$$25 = 25$$

In this example, the distributive property is used to simplify an expression. Notice that not every step is written out. As you gain experience, you can perform an operation on both sides of an equation mentally.

Example 2: Solve $-4(x - 6) = 2x$.

1. Use the distributive property to remove the grouping symbols.

$$-4(x - 6) = 2x$$

$$-4x + 24 = 2x$$

2. Add $4x$ to each side.

$$24 = 6x$$

3. Divide each side by 6.

$$4 = x$$

4. Check.

$$-4(4 - 6) = 2(4)$$

$$-4(-2) = 8$$

$$8 = 8$$

Some of the time you will be expected to write an equation from information given in the problem. The problem will describe two expressions that are equal. Write each expression in symbols and connect the expressions with the equal sign ($=$). In many problems, the word *is* indicates the $=$ symbol.

Example 3: The product of a number and 6 is 44 more than twice the number.
What is the number?

1. Write an equation. The word *is* represents the equal sign.

$$6x = 44 + 2x$$

2. Subtract $2x$ from both sides.

$$4x = 44$$

3. Divide both sides by 4.

$$x = 11$$

4. Check.

$$6(11) = 44 + 2(11)$$

$$66 = 44 + 22$$

$$66 = 66$$

The number described in the problem is **11**.

NOTE: Subtraction and division operations must be written in the order indicated by the words. "The difference between x and y " must be written $x - y$, not $y - x$. "The quotient of x and y " must be written $\frac{x}{y}$, not $\frac{y}{x}$.

Practice 1.2

A. Solve for the variable in each equation.

1. $3x - 20 = 130$
2. $2y - 8 = -3y - 18$
3. $6m = 14m - 16$
4. $2x + 5 + 6x = -27$
5. $5y + 3(y + 2) = 54$
6. $17 - 4z + 2z = 13$
7. $6m - 4 = m + 11$
8. $35 = x + 7 + 6x$
9. $5p - 2 = 6p - 9$
10. $50 = 3(s + 16) - 2(s - 2)$
11. $\frac{5(2x - 10)}{2} + 14 = 19$
12. $3(3 + r) = 2r + 4$
13. $5y = 2y + 22 + y$
14. $38 = 5(2b - 3) + 3b + 1$
15. $-5 - x = 2x - (4x + 6)$
16. $\frac{3h}{2} = 30$
17. $4(3 + 2x) + 8 = 92$
18. $-5(3 - z) = z + 1$
19. $10 - 3b + 3 = -1 + (b + 2)$
20. $5n + 8 - n = 6(n - 1)$

B. Choose the one best answer to each question.

21. Three times a number increased by 9 is 15 less than six times the number. Let x = the unknown number. Which of the following equations could be used to find the value of x ?
 - A. $3(9x) = 6(15x)$
 - B. $3x(9) = 6x - 15x$
 - C. $3x + 9 = 15 - 6x$
 - D. $3x + 9 = 6x - 15$
22. Dave has 500 baseball cards, which is as many as Eric and Travis have combined. Eric has three times as many cards as Travis has.

Dave	Eric	Travis
500	$3x$	x

From the information, you can write the equation $3x + x = 500$. How many cards does Eric have? (*Hint: Solve for x . Then find how many cards Eric has.*)

 - A. 150
 - B. 250
 - C. 350
 - D. 375
23. The difference of four times a number and 7 is 15 plus the quotient of the number and 3. Which of the following equations could be used to find the value of x ?
 - A. $4x - 7 = \frac{x}{3} + 15$
 - B. $7 - 4x = \frac{x}{3} + 15$
 - C. $7 - 4x = \frac{3}{x} + 15$
 - D. $4x - 7 = \frac{3}{x} + 1$
24. Kim earned x dollars at his part-time job on Friday. His wife earned \$12 more than twice Kim's pay ($2x + 12$). Together, they earned \$174. How much did Kim earn on Friday? (*Hint: Use the equation $x + (2x + 12) = \$174$*)
 - A. \$54
 - B. \$87
 - C. \$108
 - D. \$120

Answers and explanations begin on page 676.

Lesson 2: Equation Word Problems

Key Ideas

- To solve a word problem, decide what quantity will be represented by x . Then write expressions for the other quantities in terms of x .
- Write and solve an equation. Answer the question asked in the problem.

Algebra problems describe how several numbers are related. One number is the unknown, which you will represent with a variable. Using the relationships described in the problem, you can write an equation and solve for the variable.

Example 1: There are twice as many women as men in a class on auto repair. If there are 24 students in the class, how many are women?

1. Express the numbers in the problem in terms of the same variable. Let x represent the number of men. Since there are twice as many women, let $2x$ represent the number of women.
2. Write and solve an equation. The total number of men and women is 24, so $x + 2x = 24$. Solve: $x + 2x = 24$
 $3x = 24$
 $x = 8$

Since $x = 8$, $2x = 2(8) = 16$. There are 8 men and **16 women** in the class.

Consecutive numbers are numbers that follow in counting order. For example, 1, 2, and 3 are consecutive numbers. The numbers 2, 4, and 6 are consecutive even numbers, and 1, 3, and 5 are consecutive odd numbers.

Example 2: The sum of three consecutive numbers is 105. What is the greatest of the three numbers?

1. Let x represent the first number and $x + 1$ and $x + 2$ represent the other numbers.
2. Write an equation and solve: $x + (x + 1) + (x + 2) = 105$
 $3x + 3 = 105$
 $3x = 102$
 $x = 34$
3. Find the answer. The variable x represents the first number in the sequence, so the three numbers are 34, 35, and 36. The problem asks for the greatest number, which is **36**.

You may need to use the difference between numbers to write equations.

Example 3: The ticket prices for a play are \$12 for adults and \$8 for children. One evening, the box office sold 200 tickets. If the total box office receipts were \$2240, how many adult tickets were sold?

1. Let x represent the number of adult tickets. Since 200 tickets were sold, the number of children's tickets sold can be written as $200 - x$.

GED® TEST TIP

When you find x , you may not be finished solving a problem. Always reread the question. As shown in Examples 1 and 2, you may need to use the value of x to calculate the solution in a multi-step problem.

2. Multiply each term by the cost for that type of ticket. Set the total equal to \$2240, and solve for x .

$$\begin{aligned} 12x + 8(200 - x) &= 2240 \\ 12x + 1600 - 8x &= 2240 \\ 4x + 1600 &= 2240 \\ 4x &= 640 \\ x &= 160 \end{aligned}$$

3. There were **160 adult tickets** sold.

Practice 2

A. Solve.

- Two houses are for sale on the same street. The second house has 1000 square feet less than twice the square feet of the first house. Together the houses have 4400 square feet. What is the square footage of the first house?
- Julia has 24 coins in her pocket. The coins are either dimes or quarters. The total value of the coins is \$4.50. How many coins are dimes? (*Hint: The value of the dimes is $0.10x$, and the value of the quarters is $0.25(24 - x)$.*)
- The Bulldogs won twice as many games as they lost. If they played a total of 36 games, how many did they win? (There were no tied games.)
- The sum of four consecutive even numbers is 212. What is the third number? (*Hint: Let x = the first number, $x + 2$ = the second number, $x + 4$ = the third, and so on.*)
- A children's store is selling pants for \$6 each and shirts for \$4. Brenda bought 13 items and paid \$62. How many shirts did she buy?
- The sum of three consecutive numbers is 180. What is the least number in the series?
- In a month Andrew spends twice as much on rent as he does on food for his family. Last month, he spent \$1650 on rent and food. How much did he spend on rent?
- George spends four times as much time helping customers as he does stocking shelves. Last week, he spent 35 hours on the two tasks. How many hours were spent helping customers?

B. Choose the one best answer to each question.

- Sylvia scored 10 points better than Wiley on their science exam. Greg scored 6 points less than Wiley. Altogether, the students earned 226 points. How many points did Sylvia earn?
 - 74
 - 78
 - 84
 - 94
- Two adults and four children paid \$48 to get into the fair. A child's ticket is \$6 less than an adult's ticket. What is the cost of an adult's ticket?
 - \$15
 - \$12
 - \$9
 - \$6
- Jenny is four times as old as her niece Tina. In 12 years, Jenny will be only twice as old as Tina. The chart shows expressions for Tina and Jenny's ages now and in 12 years.

	Jenny's Age	Tina's Age
Now	$4x$	x
In 12 Years	$4x + 12$	$x + 12$

How old is Tina now?

 - 4
 - 6
 - 8
 - 12

Answers and explanations begin on page 676.

Lesson 3: Inequalities

An **inequality** is a mathematical statement that connects two unequal expressions. The inequality symbols and their meanings are as follows:

$>$ greater than	$\geq$ greater than or equal to
$<$ less than	$\leq$ less than or equal to

Key Ideas

- Use $<$, $>$, $\leq$, and $\geq$ to show the relationship between two unequal expressions.
- The symbol always points to the smaller expression.
- Use inverse operations to solve inequalities. Reverse the inequality sign if you multiply or divide by a negative number.

An inequality is solved much as is an equation. Use inverse operations to isolate the variable.

Example 1: Solve for x in the inequality $3x + 2 < 8$.

1. Subtract 2 from both sides.

$$3x + 2 < 8$$

$$3x < 6$$
2. Divide both sides by 3.

$$x < 2$$

The solution $x < 2$ states that any number less than 2 makes the inequality true. Check by substituting 1 (a number less than 2) for x : $3(1) + 2 < 8$, which simplifies to $5 < 8$, a true statement.

There is one important difference between solving equations and inequalities. Whenever you multiply or divide both sides of an inequality by a negative number, you must reverse the inequality symbol.

Example 2: Solve for n in the inequality $-2n - 5 \geq 3$.

1. Add 5 to both sides to remove -5 from the left side of the equation.

$$-2n - 5 + 5 \geq 3 + 5$$

$$-2n \geq 8$$
2. Divide both sides by -2 and *reverse the inequality symbol*.

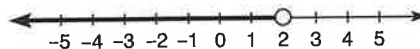
$$\frac{-2n}{-2} \geq \frac{8}{-2}$$

$$n \leq -4$$

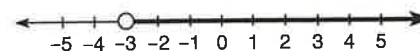
Check your work by substituting a number that is less than or equal to -4 into the original inequality. Here -5 is used for n . Since $5 \geq 3$ is a true statement, the answer is correct.

When an inequality contains a variable, there are usually several numbers that make the inequality true. For that reason, we often graph the solution. In the examples below, a closed dot means that the number is included in the solution set. An open dot means the number is not included.

Examples: $x < 2$



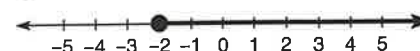
$x > -3$



$x \leq 1$



$x \geq -2$



GED® TEST TIP

Avoid the most common mistake with inequalities: reversing the symbol accidentally. Always check your work by substituting any number that fits the solution into the original inequality.

A **compound inequality** combines two inequalities. To solve a compound inequality, separate the inequalities and solve both. Then combine the solutions.

Example 3: Solve $3x + 4 < 5x < 16 + x$.

- | | | |
|---|---------------|---------------|
| 1. Write two inequalities and solve each separately. | $3x + 4 < 5x$ | $5x < 16 + x$ |
| | $4 < 2x$ | $4x < 16$ |
| 2. Write the result as a compound inequality. $2 < x < 4$ | $2 < x$ | $x < 4$ |

In other words, any quantity that is greater than 2 *and* less than 4 will make the compound inequality true.

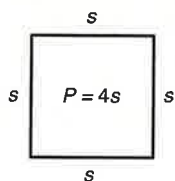
Practice 3

A. Solve.

- | | | |
|-----------------------------|------------------------------|-----------------------------------|
| 1. $3x - 7 > 5$ | 9. $\frac{x-2}{3} > 2x + 11$ | 17. $6 \leq 3(x + 3)$ |
| 2. $13 < 2x - 1$ | 10. $6x < 5x + 2$ | 18. $\frac{4x}{3} > 8x - 20$ |
| 3. $4 + 2x \leq -2$ | 11. $x + 6 \leq 8x - 15$ | 19. $x - 2 < \frac{2x + 6}{4}$ |
| 4. $\frac{4 + x}{5} \leq 8$ | 12. $5x + 14 > 2 + 7x$ | 20. $x \geq 4x - 9$ |
| 5. $2(x + 3) < 4$ | 13. $13x - 7 > 25 - 3x$ | 21. $30 \geq 5(x + 4) \geq 10$ |
| 6. $3 + 9x \geq 4(x + 7)$ | 14. $x - 6 < 2(x + 2)$ | 22. $-7x > -2(x + 15) < 10$ |
| 7. $-4(x + 2) < 24$ | 15. $-5 + 3x \geq 4(3x - 8)$ | 23. $3 < 5x - 27 < 53$ |
| 8. $-2x + 9 < 1$ | 16. $36 > 4(x - 12)$ | 24. $22 \leq 6x - 2 \leq 4x + 16$ |

B. Choose the one best answer to each question.

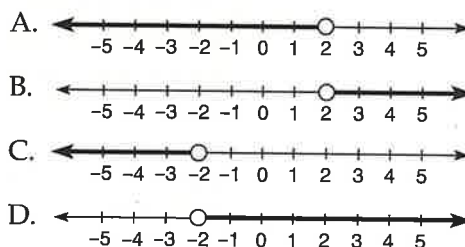
25. The perimeter of a square can be found using the formula $P = 4s$, where s is one side of the square.



The perimeter of a square is less than or equal to 64 inches. Which of the following represents the possible measures of the side of the square in inches?

- A. $s \leq 16$
 B. $s \geq 16$
 C. $s \geq 8$
 D. $s \leq 64$

26. Three added to the product of -4 and a number (x) is less than 5 added to the product of -3 and the number. Which of the following is a graph of the solution set of x ?



Answers and explanations begin on page 677.

Lesson 4: Quadratic Equations

Key Ideas

- A quadratic equation can have two solutions.
- Set the quadratic equation equal to 0 and factor.
- Find the values for x that will make each factor equal 0.

A **quadratic equation** contains a squared variable, for example, $x^2 - 3x = 4$. One way to solve quadratic equations is by factoring. This is the simplest method and the one you should use on the GED® *Mathematical Reasoning Test*. When you factor an expression, you find the terms that divide evenly into the expression.

Example 1: Factor the expression $15x^2 + 9x$.

1. Look for a term that divides evenly into both $15x^2$ and $9x$. Both terms can be divided by $3x$.
 $15x^2 \div 3x = 5x$
 $9x \div 3x = 3$
2. Factor out $3x$. Write the terms as factors.
 $15x^2 + 9x = 3x(5x + 3)$

Study Example 2 to learn how to multiply factors with more than one term.

Example 2: Multiply $(2x + 3)(x - 4)$.
(factor) (factor)
four terms

1. Multiply each term in the first factor by each term in the second factor.
 $2x \cdot x = 2x^2$
 $2x \cdot (-4) = -8x$
 $3 \cdot x = 3x$
 $3 \cdot (-4) = -12$
 $2x^2 + (-8x) + 3x + (-12) =$
 $2x^2 - 5x - 12$
2. Combine the results.

This method of multiplying factors is called the FOIL method. The letters in FOIL stand for First, Outer, Inner, and Last. Use the word FOIL to make sure you have performed all the necessary operations.

You factor a quadratic equation to solve it. A quadratic equation may have two solutions. Find values that make the factors equal to 0. Since a number multiplied by 0 is 0, each of the values is a solution.

Example 3: Solve $x^2 - 3x = 4$.

1. Set the equation equal to 0 by subtracting 4 from both sides.
 $x^2 - 3x = 4$
 $x^2 - 3x - 4 = 0$
2. Factor by trial and error. Think: What factors of the last term, -4 , when added, will equal -3 , the number part of the middle term?
 $(x \quad)(x \quad) = 0$
 $(x + 1)(x - 4) = 0$
 $-4 \cdot 1 = -4$ and $-4 + 1 = -3$

3. If either one of the factors equals 0, then the product of the factors will be 0. Set each factor equal to 0 and solve for x .
 $x + 1 = 0$
 $x = -1$
 $x - 4 = 0$
 $x = 4$

The solutions to the quadratic equation are **-1 and 4** .

GED® TEST TIP

Trial and error can help you factor quadratic equations. The more you practice, the better you will get at solving them.

Practice 4

A. Multiply.

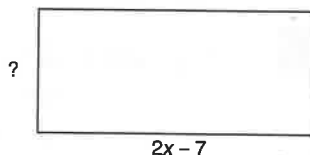
- | | | |
|-----------------|-------------------|--------------------|
| 1. $(x+4)(x+2)$ | 6. $(2x+1)(x-2)$ | 11. $(x-6)(x+5)$ |
| 2. $(x-3)(x+5)$ | 7. $(x-9)(x-5)$ | 12. $(x-10)(x-3)$ |
| 3. $(x-1)(x+4)$ | 8. $(x+1)(3x-2)$ | 13. $(2x+1)(2x+2)$ |
| 4. $(x-6)(x-3)$ | 9. $(x-2)(x+7)$ | 14. $(x+9)(x-4)$ |
| 5. $(x+8)(x-2)$ | 10. $(3x+8)(x+2)$ | 15. $(x-5)(x-5)$ |

B. Factor each expression.

- | | | |
|---------------------|----------------------|----------------------|
| 16. $x^2 + 4x + 3$ | 21. $x^2 - x - 12$ | 26. $2x^2 + 5x - 3$ |
| 17. $x^2 + 4x - 5$ | 22. $x^2 + 2x - 35$ | 27. $2x^2 - 8x - 10$ |
| 18. $x^2 + 8x + 12$ | 23. $x^2 - 12x + 36$ | 28. $x^2 + 5x - 50$ |
| 19. $x^2 - x - 6$ | 24. $x^2 - 6x - 7$ | 29. $4x^2 + 4x - 3$ |
| 20. $x^2 + 5x - 14$ | 25. $x^2 + 4x - 32$ | 30. $x^2 + x - 56$ |

C. Choose the one best answer to each question.

31. What are two solutions for the equation $x^2 - x = 20$?
- A. -4 and 5
B. 4 and -5
C. -10 and 2
D. -2 and 10
34. The area of a rectangle is found by multiplying the length by the width. In the rectangle below, the area of the rectangle is equal to the expression $2x^2 - 27x + 70$.



32. For which of the following equations is $x = -4$ a solution?
- A. $2x^2 - 8 = 0$
B. $x^2 - 8x + 64 = 0$
C. $x^2 - 2x - 15 = 0$
D. $2x^2 + 2x - 24 = 0$
33. What is the only positive solution for the equation $2x^2 - 7x - 30 = 0$?
- A. 5
B. 6
C. 7
D. 8

An expression equal to the length is shown on the diagram. Which of the following expressions is equal to the width of the rectangle?

- A. $x + 10$
B. $x - 10$
C. $2x + 10$
D. $2x - 10$

Answers and explanations begin on page 677.

Lesson 5: Algebra Problem Solving

Key Ideas

- Guess-and-check is a strategy for answering complicated multiple-choice problems.
- Choose an answer from the options and try the value in the problem. If it works, you have found the correct answer.

One strategy that can help you save time on the more complicated algebra problems is **guess-and-check**. Guess-and-check means selecting one of the answer choices and trying that value in the problem. If you guess correctly, you can move on to the next question. If not, guess again. Guess-and-check is a good strategy for problems involving quadratic equations and expressions.

Example 1: Which of the following is a solution for the equation $2x^2 - 12 = 2x$?

- A. 4
- B. 3
- C. 0
- D. -1

To solve the problem, you would have to rewrite the equation so that the quadratic expression equals zero, factor the expression, and solve.

Instead, substitute each answer choice into the equation.

Option (A):	$2x^2 - 12 = 2x$	Option (B):	$2x^2 - 12 = 2x$
	$2(4)^2 - 12 = 2(4)$		$2(3)^2 - 12 = 2(3)$
	$20 \neq 8$		$6 = 6$

Option (B) 3 makes the equation true.

Guess-and-check can also save time when writing an equation seems difficult.

Example 2: Terry is ten years older than his brother Tomas. Twenty years ago, Terry was twice as old as Tomas. How old is Terry now?

- A. 25
- B. 30
- C. 40
- D. 45

Instead of writing an equation, try each age in the answer choices for Terry.

- A. If Terry is 25 now, Tomas is 15. Twenty years ago, Tomas would not have been born.
- B. If Terry is 30 now, Tomas is 20. Twenty years ago, Tomas would have been 0 years old, and Terry would have been 10.
- C. If Terry is 40 now, Tomas is 30. Twenty years ago, Tomas would have been 10, and Terry would have been 20, which is twice as old as 10.

Therefore, **option (C) 40** is correct.

GED® TEST TIP

Use your judgment. Do not use guess-and-check if you can quickly see how to solve the problem. Trying each answer choice may take longer than simply solving the problem.